EXPERIMENT 14 PPS

14.1.1//Complex number addition using class

Ans: ALGORITHM;

Step 1: Start

Step 2: Define a class Complex with attributes:

- real
- imag

Step 3: Define method initComplex()

- Read two integers from user input
- Assign them to real and imag

Step 4: Define method sum(c1, c2)

- Set real = c1.real + c2.real
- Set imag = c1.imag + c2.imag

Step 5: Define method display()

- If imag >= 0, print "real + imag i"
- Else, print "real - |imag| i"

Step 6: Create three objects:

- c1, c2, c3 of class Complex

Step 7: Call initComplex() for c1

- Input first complex number

Step 8: Call initComplex() for c2

- Input second complex number

Step 9: Call sum(c1, c2) using c3

- Store result in c3

Step 10: Call display() using c3

- Display the result

Step 11: Stop

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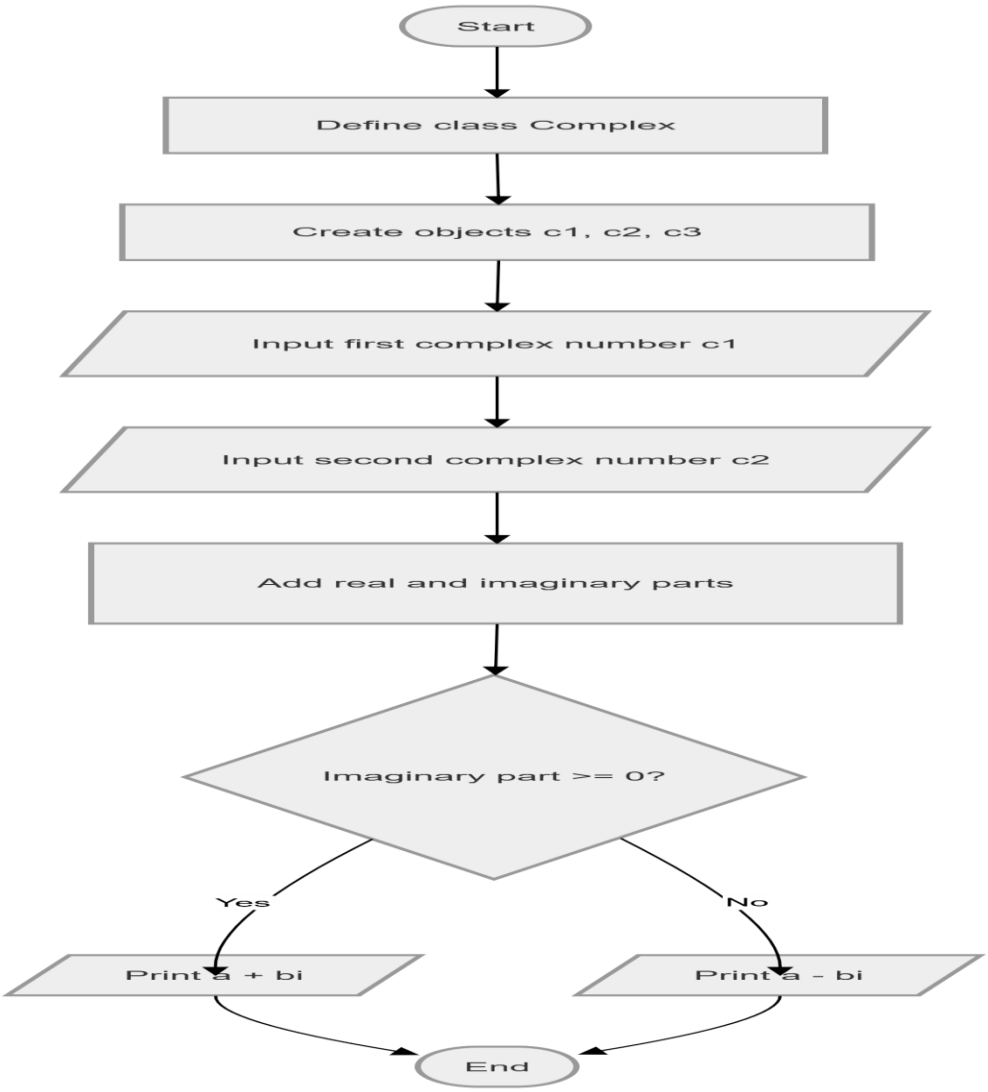
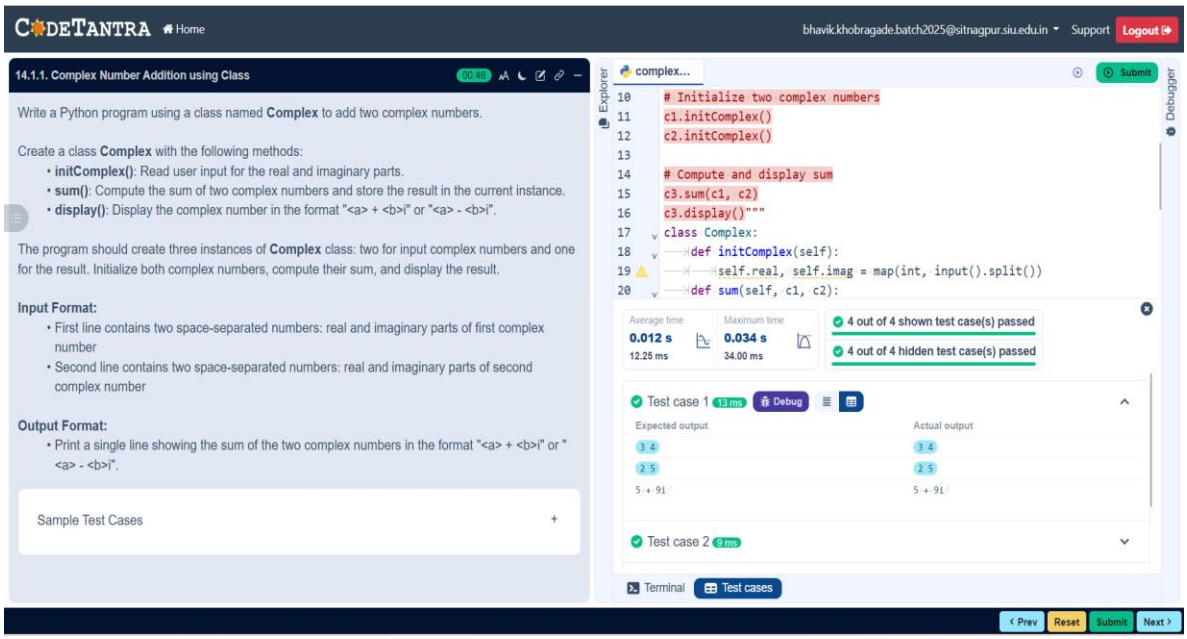

Start
Define class Complex
Create objects c1, c2, c3
Input first complex number c1
Input second complex number c2
Add real and imaginary parts
Imaginary part >= 0?
Yes
No
Print a + bi
Print a - bi
End

CODETANTRA: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14.1.1. Complex Number Addition using Class
Write a Python program using a class named Complex to add two complex numbers.
Create a class Complex with the following methods:
• initComplex(): Read user input for the real and imaginary parts.
• sum(): Compute the sum of two complex numbers and store the result in the current instance.
• display(): Display the complex number in the format "<a> + <b>i" or "<a> - <b>i".
The program should create three instances of Complex class: two for input complex numbers and one for the result. Initialize both complex numbers, compute their sum, and display the result.
Input Format:
• First line contains two space-separated numbers: real and imaginary parts of first complex number
• Second line contains two space-separated numbers: real and imaginary parts of second complex number
Output Format:
• Print a single line showing the sum of the two complex numbers in the format "<a> + <b>i" or "<a> - <b>i".
Sample Test Cases
complex...
Submit
# Initialize two complex numbers
c1.initComplex()
c2.initComplex()
# Compute and display sum
c3.sum(c1, c2)
c3.display()"""
class Complex:
def initComplex(self):
self.real, self.imag = map(int, input().split())
def sum(self, c1, c2):
Average time
0.012 s
12.25 ms
Maximum time
0.034 s
34.00 ms
4 out of 4 shown test case(s) passed
4 out of 4 hidden test case(s) passed
Test case 1
Debug
Expected output
Actual output
Test case 2
Terminal
Test cases
Prev
Reset
Submit
Next